CS 237: PROBABILITY IN COMPUTING

INSTRUCTORS: NATHAN KLEIN AND TIAGO JANUARIO

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LECTURE 3

Last time
- Symmetry
- Probability function
- Probability axioms

Today
- Why do you need the axioms?
- Tree diagrams
- Analysis of the Monty Hall problem
- Dice game

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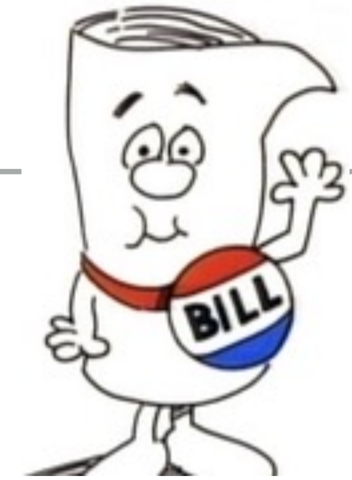
GET READY FOR VOTING!

- Students from A1 section:
  1. Enter this URL: https://app.tophat.com/e/779973
  2. Login as needed using your Top Hat account info.
  3. If asked about enrolling, click *Enroll*.
- Students from A2 section:
  1. Enter this URL: https://app.tophat.com/e/544135
  2. Login as needed using your Top Hat account info.
  3. If asked about enrolling, click *Enroll*.



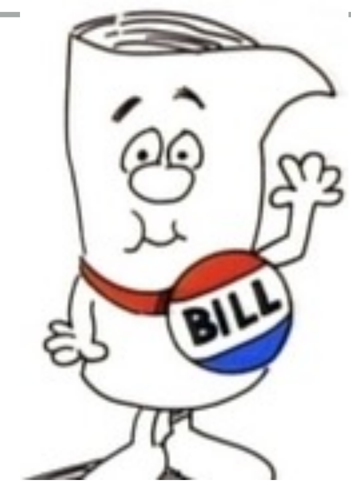
I'M JUST A BILL

- For a bill to come before the US president, it must be passed by both the House and the Senate
- Suppose that 40% of bills pass the House, 30% the Senate, and 50% pass at least one of the two
- What is the probability the next bill will come before the president?

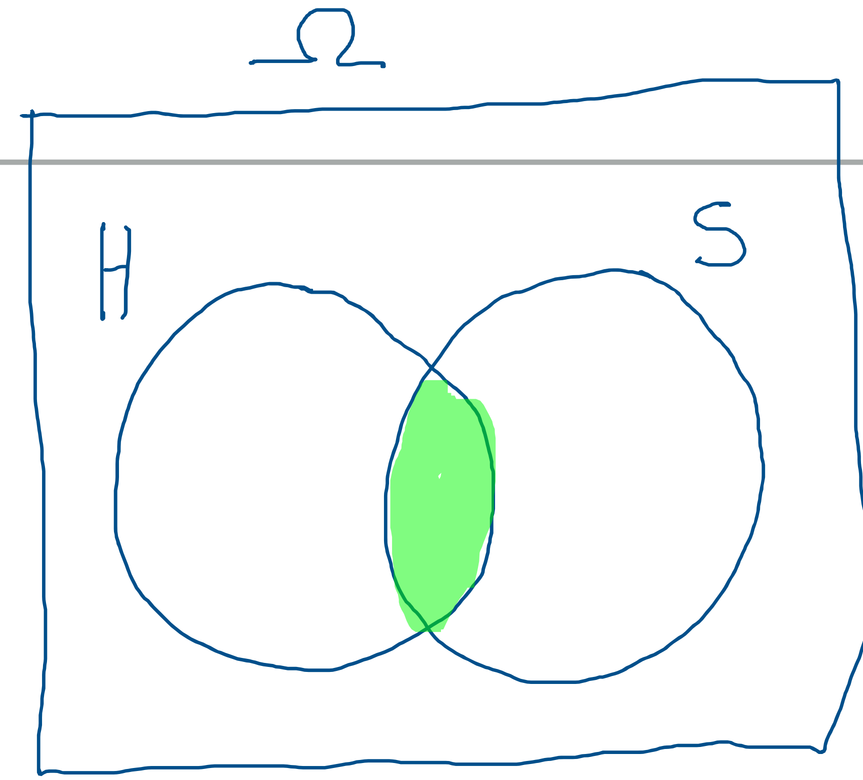
(A) 0.2 (B) 0.4 (C) 0.5 (D) 0.7

(E) None of the above

I'M JUST A BILL



- Solution:



H = bill passes the house $\rightarrow Pr(H) = 40\% = 0.4$

S = bill passes the senate $\rightarrow Pr(S) = 30\% = 0.3$

$$Pr(H \cup S) = 0.5 = Pr(H) + Pr(S) - Pr(H \cap S)$$

$$Pr(H \cap S) = 0.4 + 0.3 - 0.5 = 0.7 - 0.5 = 0.2$$

Probability that the bill passes both House and Senate

BETTING ON DICE

- We have a standard die and we roll it 4 times
- Should we bet on "at least one 6 turns up"?

Chevalier de Mere
[1607-1684]

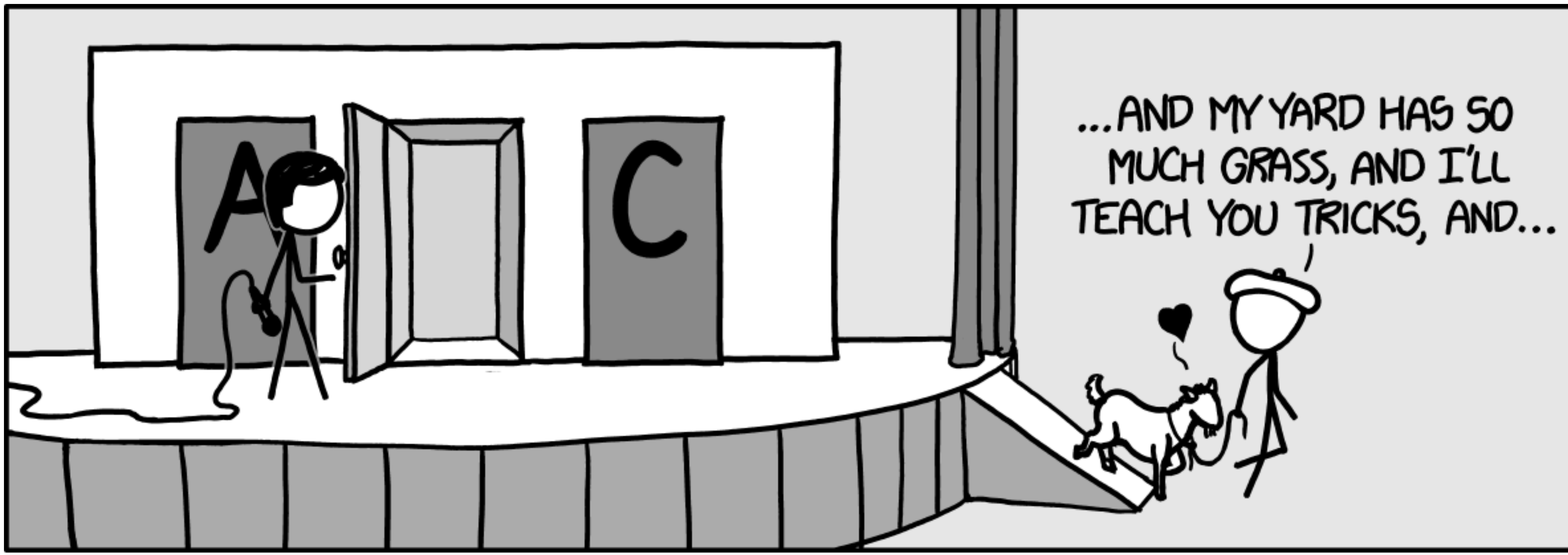
A = "at least one 6 turns up in 4 rolls of a die"

∴ think outside the box...

$\bar{A}$ = no 6 turns up in 4 rolls

$$\Pr(\bar{A}) = \left(\frac{5}{6}\right)^4 \Rightarrow \Pr(A) = 1 - \Pr(\bar{A}) = 1 - \left(\frac{5}{6}\right)^4 \approx 0.51$$

PROBABILITY AXIOMS AND WHY WE NEED THEM



https://xkcd.com/1282/

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ASK THE AUDIENCE

"Linda is 31, outspoken, and very bright. She majored in philosophy in college. As a student, she was deeply concerned with racial discrimination and other social issues, and participated in anti-nuclear demonstrations"

- Rank the likelihood of the following alternatives:

  (A) Linda is active in the feminist movement

  (B) Linda is a bank teller

  (C) Linda is a bank teller and active in the feminist movement

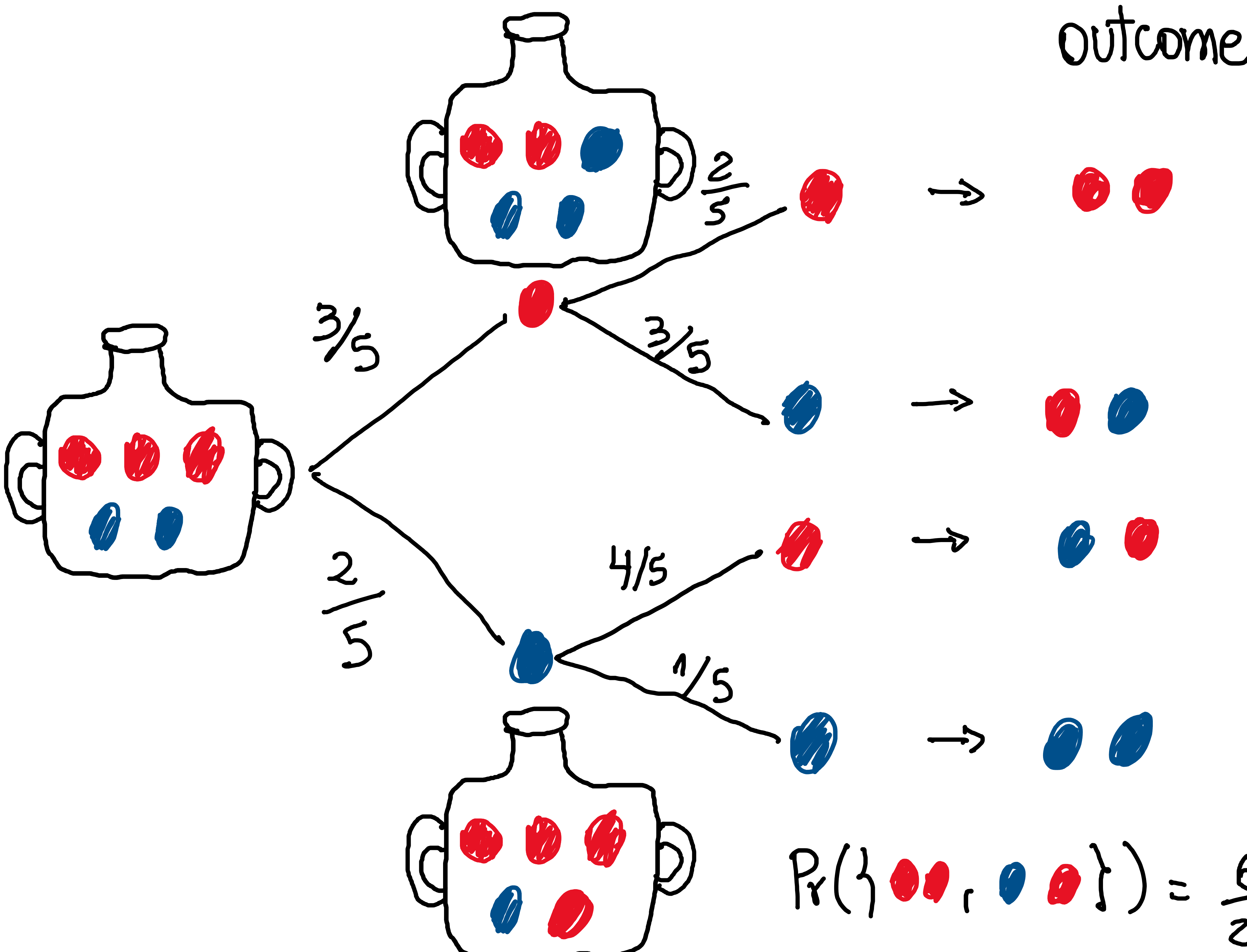
TREE DIAGRAM METHOD

- Very useful problem-solving approach
- Allows us to analyze complicated experiments
- Especially useful when there are several random choices that depend on one another

THE CLASSIC URN PROBLEM

- An urn contains 3 red balls and 2 blue balls
- Draw a ball uniformly at random
  - If the ball drawn is red, add a blue ball to the urn
  - If the ball drawn is blue, add a red ball to the urn
  - The drawn ball is not returned to the urn
- Draw a second ball uniformly at random
- Pr("second ball is red") = ?

THE CLASSIC URN PROBLEM – SOLUTION



Probability

$\frac{3}{5} \cdot \frac{2}{5} = \frac{6}{25}$

$\frac{3}{5} \cdot \frac{3}{5} = \frac{9}{25}$

$\frac{2}{5} \cdot \frac{4}{5} = \frac{8}{25}$

$\frac{2}{5} \cdot \frac{1}{5} = \frac{2}{25}.$

$\Pr(\{\ ,\ \}) = \frac{6}{25} + \frac{8}{25} = \frac{14}{25}$

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LET THE GOOD TIMES ROLL

- Blue die: 0, 0, 4, 4, 4, 4
- Green die: 1, 1, 1, 5, 5, 5
- Orange die: 2, 2, 2, 2, 6, 6
- Yellow die: 3, 3, 3, 3, 3, 3
- You get to pick a die first
- I pick one of the remaining dice
- We roll, the larger number wins
- We play 10,000 times (by computer)

Which die would you choose?

(A) Blue (B) Green

(C) Orange (D) Yellow

(E) I would ask to pick second

$\Pr(\text{Green beats Blue}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{3} = \frac{2}{3}$

LET THE GOOD TIMES ROLL

- Blue die: 0, 0, 4, 4, 4, 4
- Green die: 1, 1, 1, 5, 5, 5
- Orange die: 2, 2, 2, 2, 6, 6
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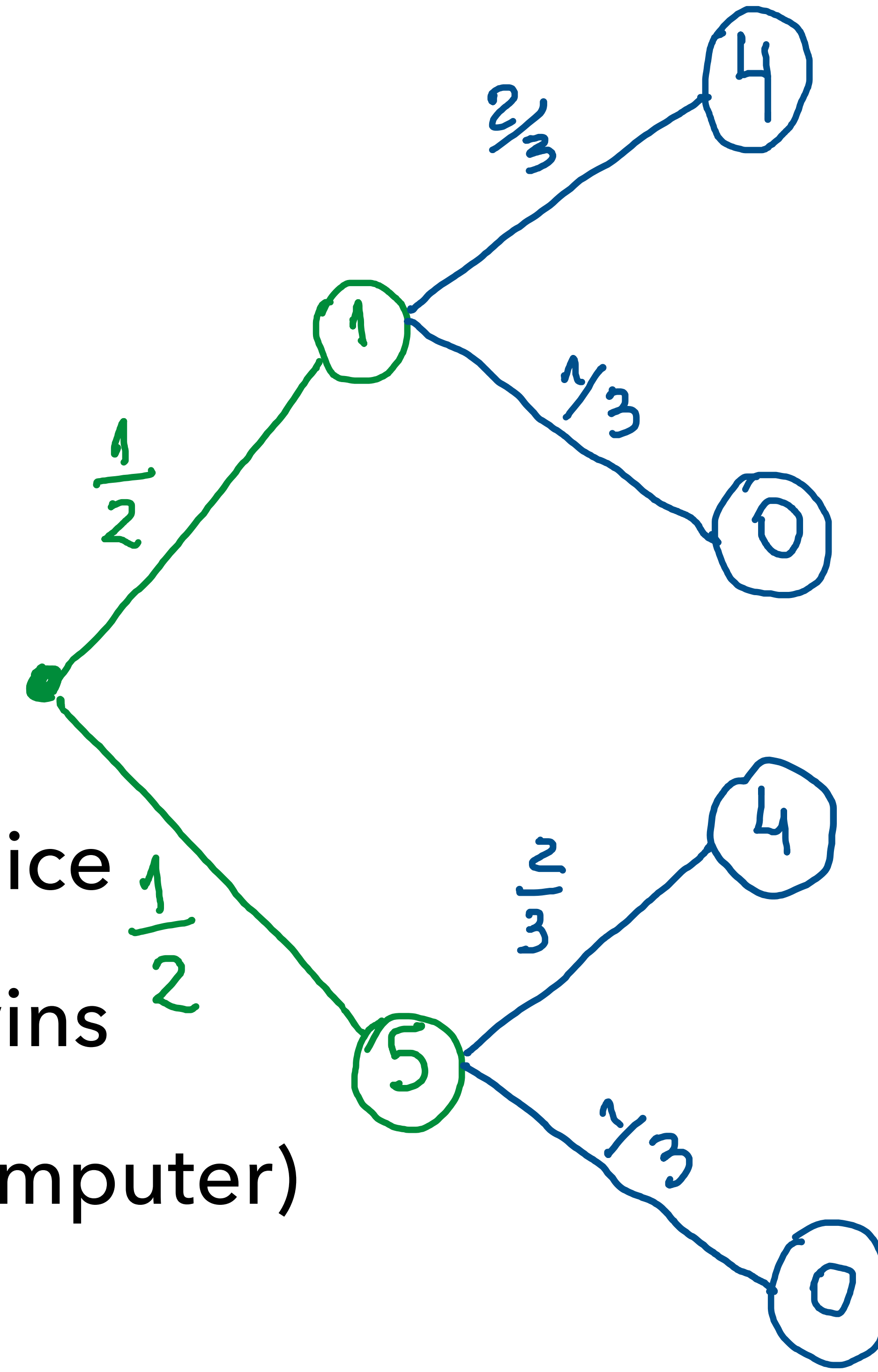

outcome	Prob.	W
(1,4)	$\frac{1}{2}\cdot\frac{2}{3}=\frac{1}{3}$	B
(1,0)	$\frac{1}{2}\cdot\frac{1}{3}=\frac{1}{6}$	G
(5,4)	$\frac{1}{2}\cdot\frac{2}{3}=\frac{1}{3}$	G
(5,0)	$\frac{1}{2}\cdot\frac{1}{3}=\frac{1}{6}$	G

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NON-TRANSITIVE DICE ANALYSIS

(Analysis on the slide)

Blue die: 0, 0, 4, 4, 4, 4

Green die: 1, 1, 1, 5, 5, 5

Orange die: 2, 2, 2, 2, 6, 6

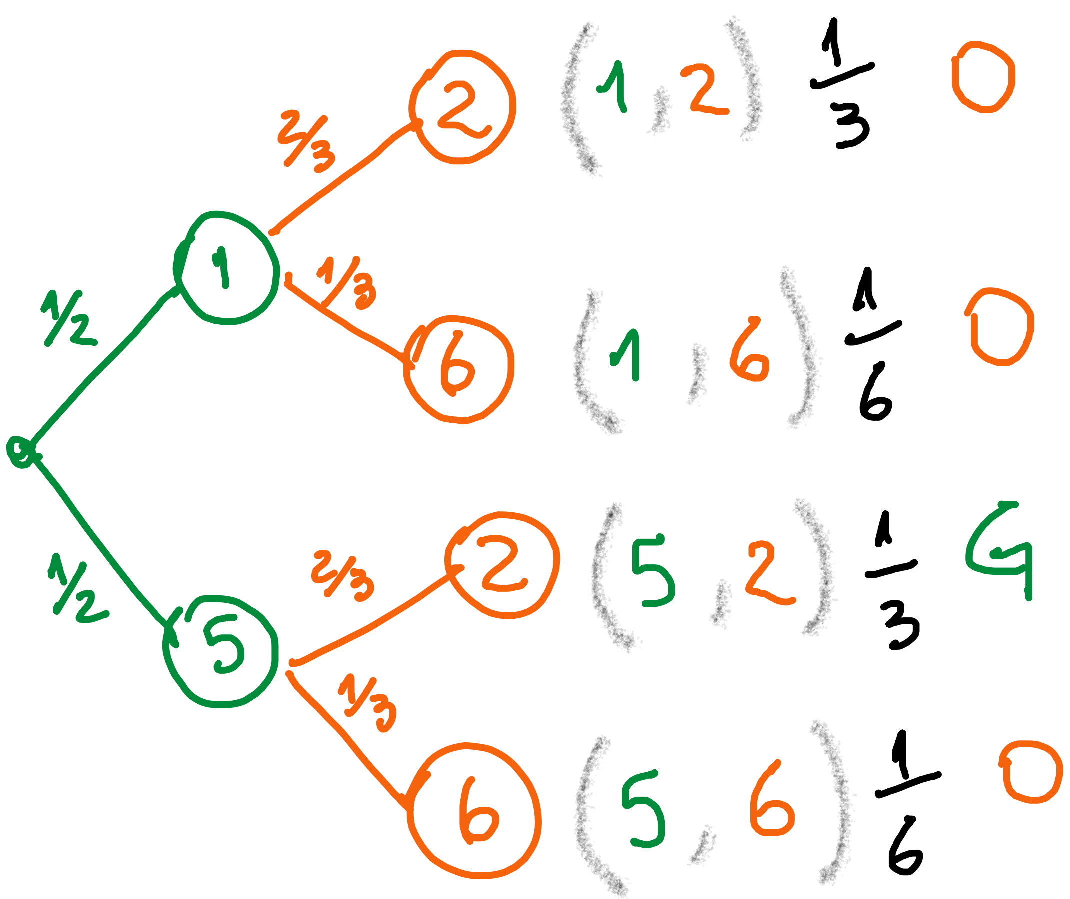
Yellow die: 3, 3, 3, 3, 3, 3

beats

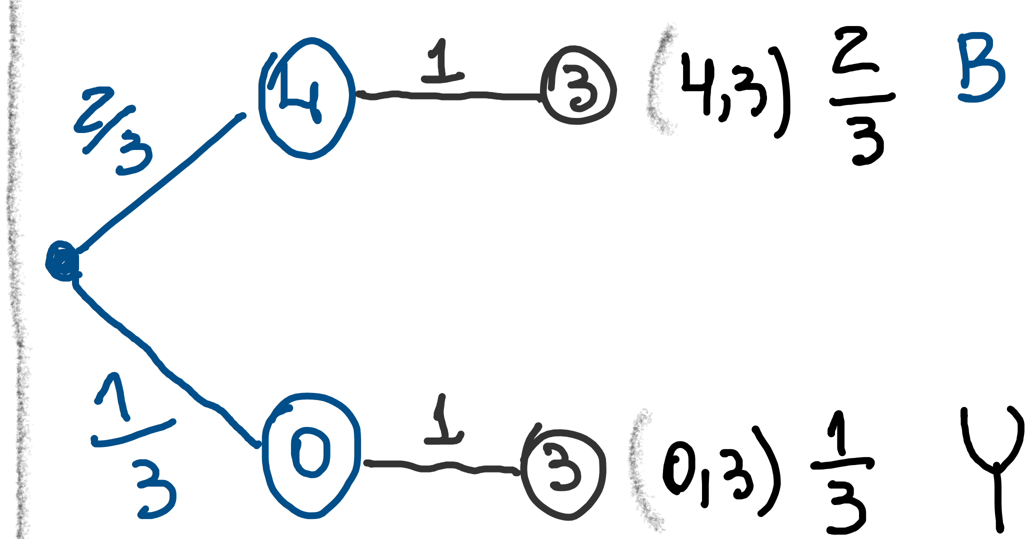
beats

beats

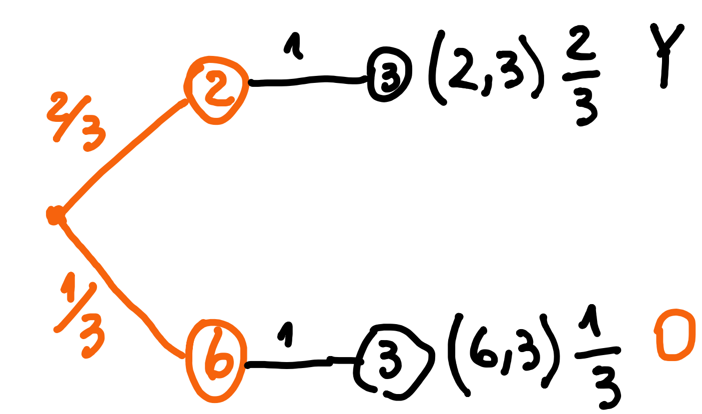
beats



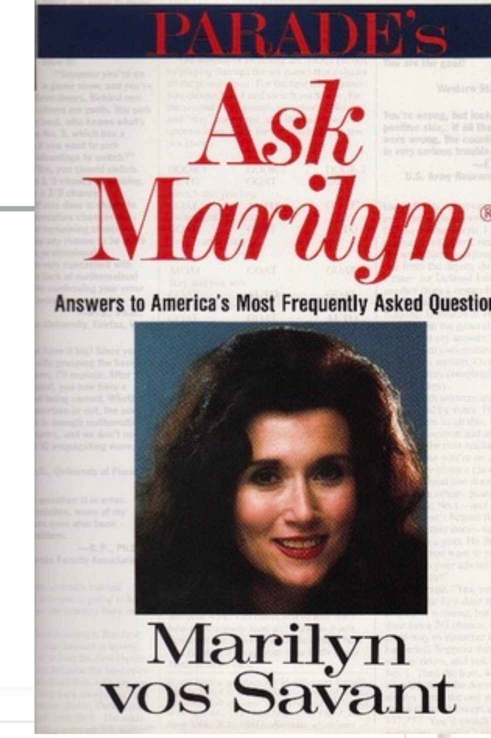
$Pr(\text{Orange beats Green}) = \frac{2}{3}$



$Pr(\text{Blue beats Yellow}) = 2/3$



$Pr(\text{Yellow beats Orange}) = \frac{2}{3}$



ASK MARLYN

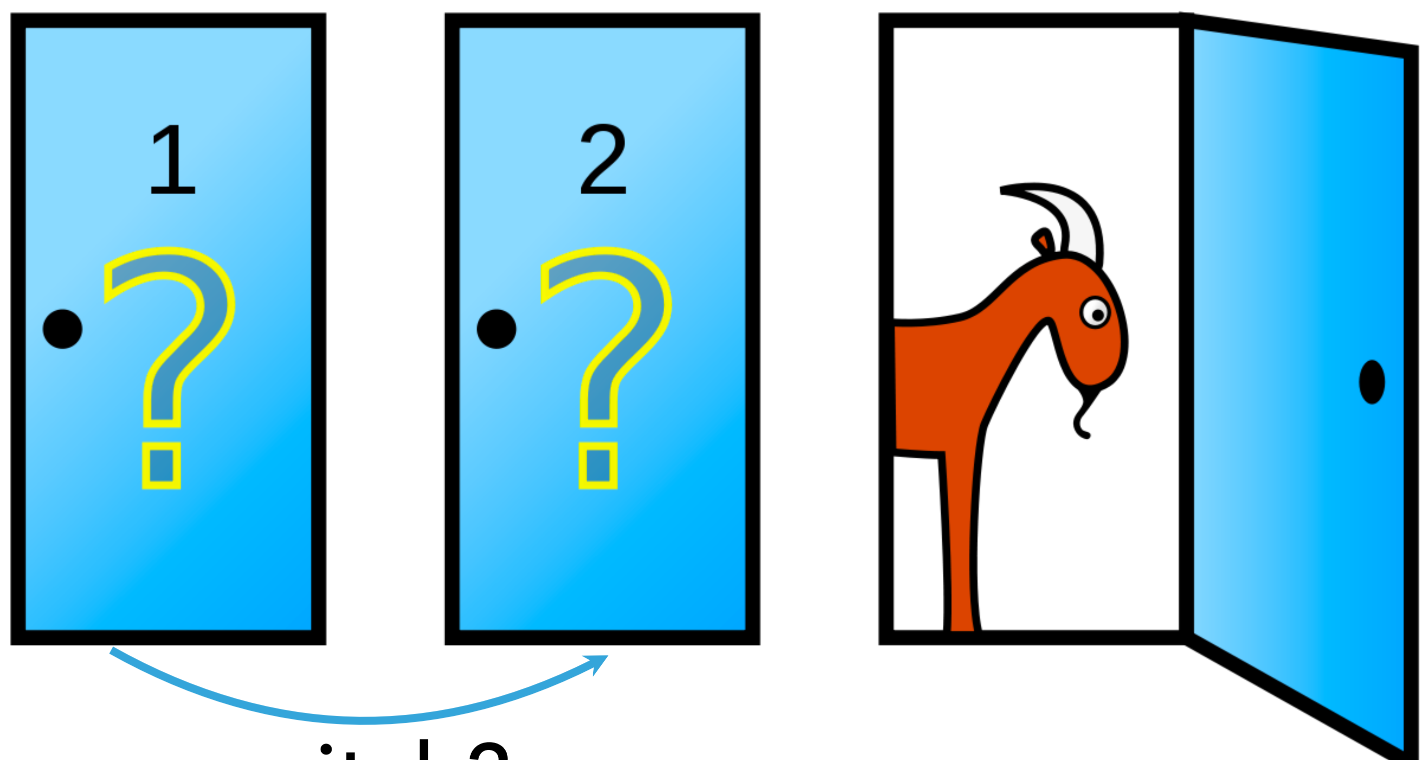
> Suppose you're on a game show, and you're given the choice of three doors. Behind one door is a car, behind the others, goats. You pick a door, say number 1, and the host, who knows what's behind the doors, opens another door, say number 3, which has a goat. He says to you, "Do you want to pick door number 2?" Is it to your advantage to switch your choice of doors?

Craig. F. Whitaker
Columbia, MD

LET'S MAKE A DEAL

contestant picks this door

host opens this door



switch?

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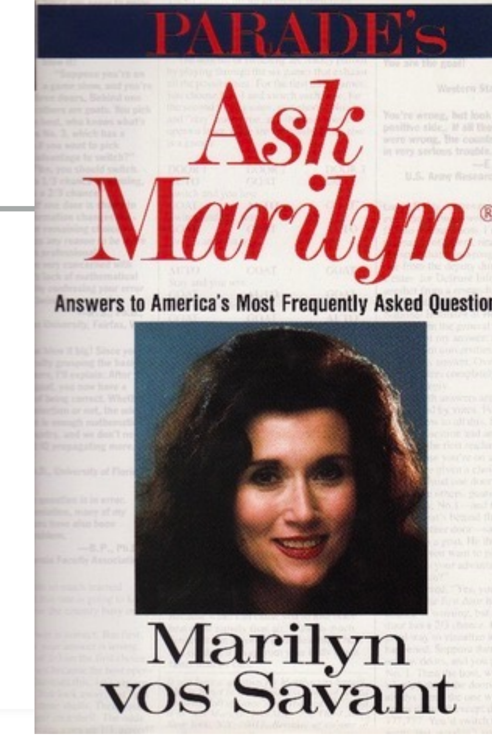
MODELING THE PROBLEM

- The car is equally likely to be behind each of the 3 doors
- After the contestant picks a door, the host opens a different door with a goat behind it and offers the contestant the choice of switching to the remaining door
- If the host has a choice of which door to open, he is equally likely to pick each of them

(A) switch (B) not switch (C) it does not matter

18

MARILYN'S ANSWER



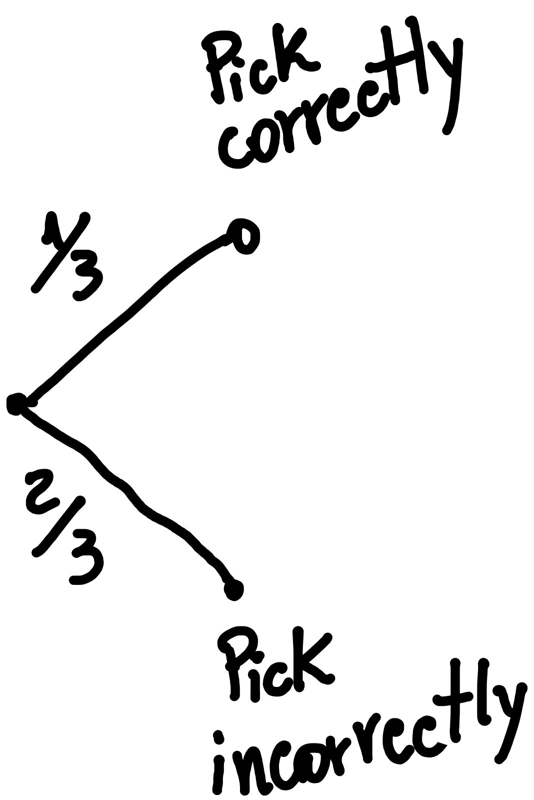
Yes; you should switch. The first door has a 1/3 chance of winning, but the second door has a 2/3 chance.

Here's a good way to visualize what happened. Suppose there are a million doors, and you pick door #1. Then the host, who knows what's behind the doors and will always avoid the one with the prize, opens them all except door #777,777. You'd switch to that door pretty fast, wouldn't you?

LET'S MAKE A DEAL

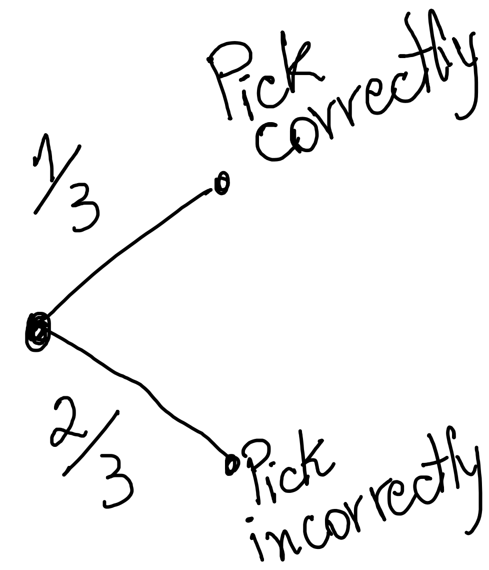
- Marilyn's proof.

Stay strategy



Outcomes	Pr
Win	1/3
lose	2/3

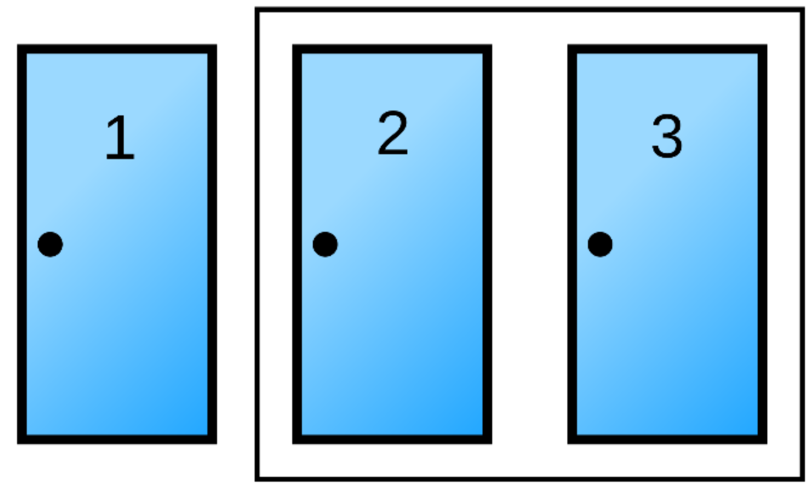
Switch strategy



Outcomes	Prob.
Win	2/3
lose	1/3

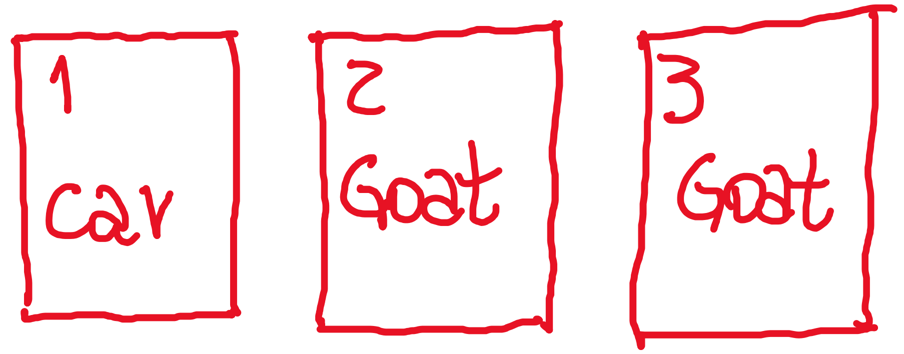
MONTY HALL PROBLEM: INTUITION

- If you pick the **correct** door initially, you win iff you stick with it.
- If you pick the **wrong** door initially, you win iff you switch.
- What is the probability you get it right initially?



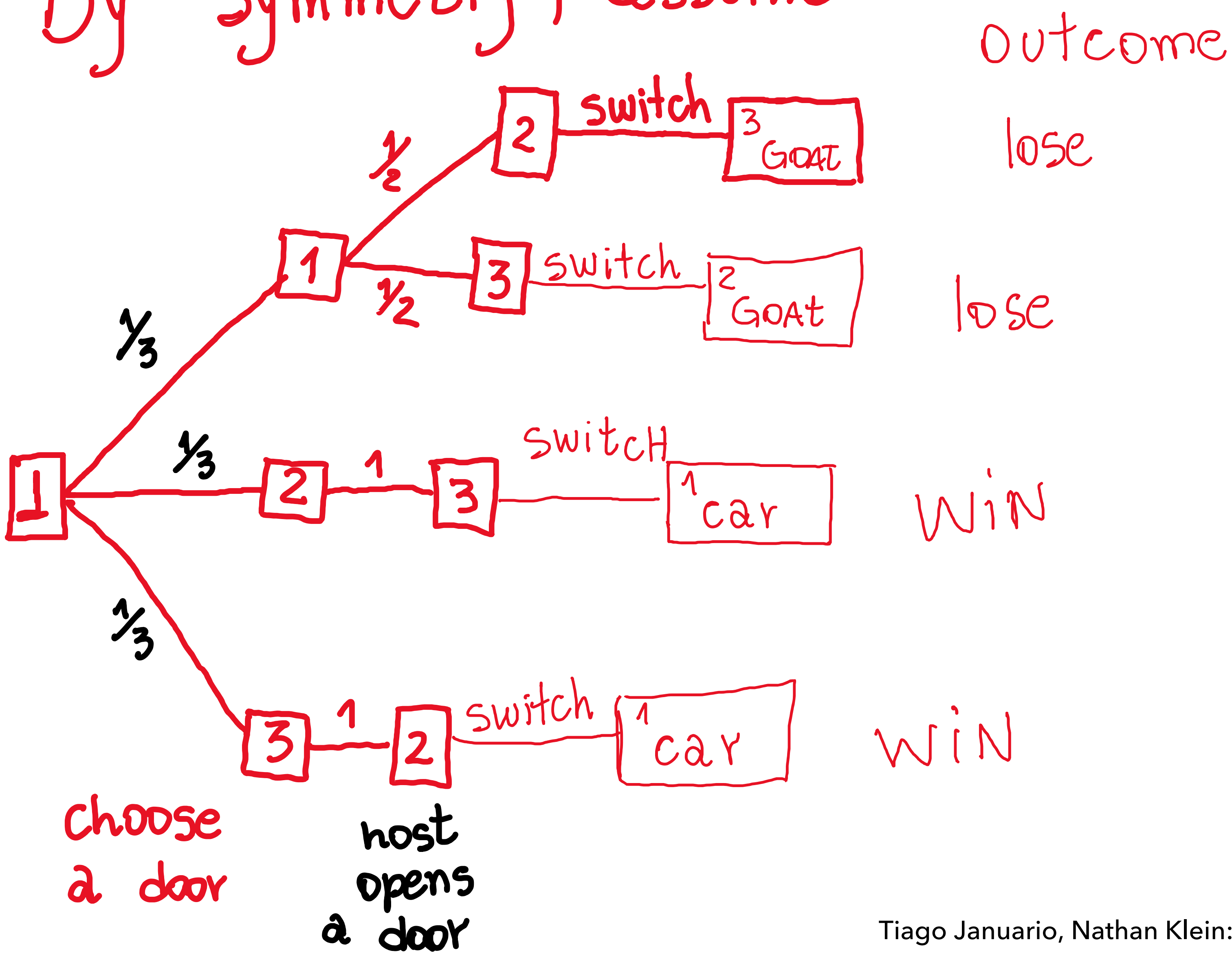
- ``Sticking with it'' player wins with probability 1/3.
- Switching player wins with probability 2/3.

LET'S MAKE A DEAL



- **Question:** should my strategy depend on *which door the host opens?* Should I sometimes switch and sometimes stay?

By symmetry, assume the car is behind door 1



Probabilities

$\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$

$\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$

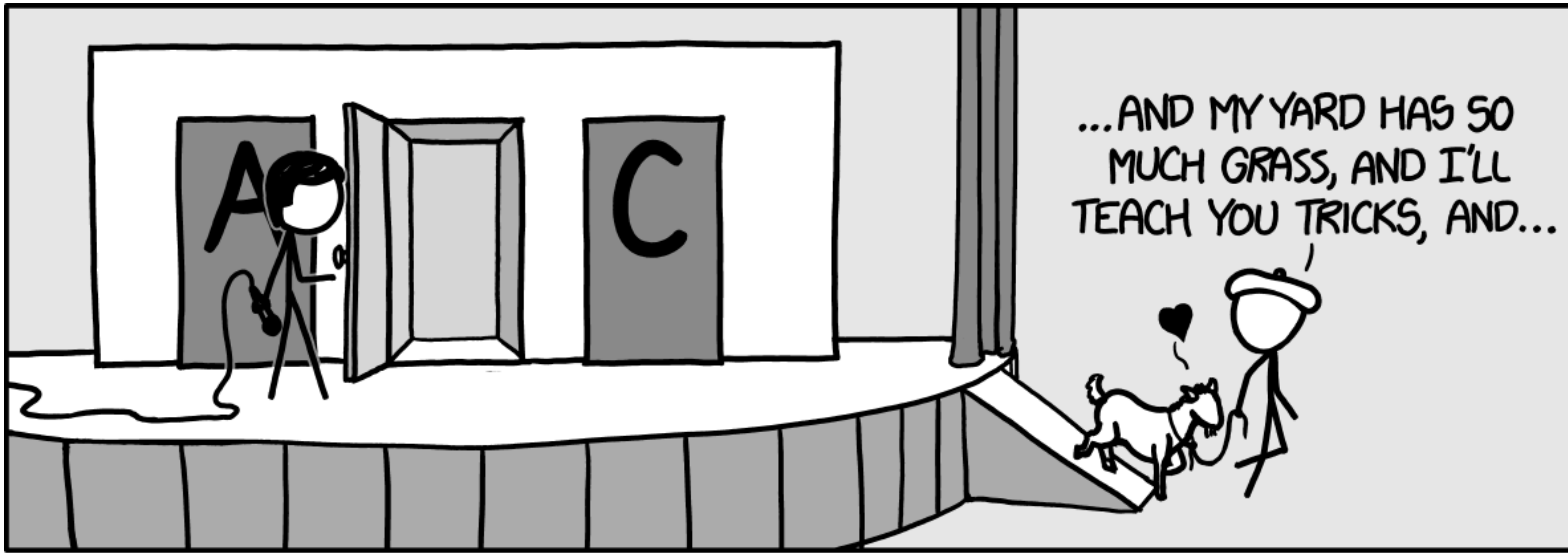
$\left.\right\}$ $Pr(Lose) = \frac{1}{3}$

$\frac{1}{3} \cdot 1 = \frac{1}{3}$

$\frac{1}{3} \cdot 1 = \frac{1}{3}$

$\left.\right\}$ $Pr(Win) = \frac{2}{3}$

THAT'S ALL FOR MONTY HALL



https://xkcd.com/1282/

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PROBABILITY IS TRICKY, EVEN FOR EXPERTS

"Many readers of vos Savant's column refused to believe switching is beneficial despite her explanation. After the problem appeared in Parade, approximately 10,000 readers, including nearly 1,000 with PhDs, wrote to the magazine, most of them claiming vos Savant was wrong. "

From the Wikipedia article on the Monty Hall problem:

https://en.wikipedia.org/wiki/Monty_Hall_problem

Since you seem to enjoy coming straight to the point, I'll do the same. You blew it! Let me explain. If one door is shown to be a loser, that information changes the probability of either remaining choice, neither of which has any reason to be more likely, to 1/2. As a professional mathematician, I'm very concerned with the general public's lack of mathematical skills. Please help by confessing your error and in the future being more careful.

–Robert Sachs, Ph.D.
George Mason University

You blew it, and you blew it big! Since you seem to have difficulty grasping the basic principle at work here, I'll explain. After the host reveals a goat, you now have a one-in-two chance of being correct. Whether you change your selection or not, the odds are the same. There is enough mathematical illiteracy in this country, and we dont need the world's highest IQ propagating more. Shame!

–Scott Smith, Ph.D.
University of Florida

You are indeed correct. My colleagues at work had a ball with this problem, and I dare say that most of them, including me at first, thought you were wrong!

–Seth Kalson, Ph.D.
Massachusetts Institute of Technology

https://www.math.ucdavis.edu/~swalcott/Monty_Hall_17Cs2018.pdf

PUTTING IT TO THE TEST

Marilyn called out for a nation-wide experiment, and many math teachers and students simulated the experiment and sent her the results.

```
# simulates a single trial of the experiment
def monty_hall_single_trial():
    winning_door = randint(1,4) # the door with the car
    contestant_door = randint(1,4) # the door chosen by contestant
    remaining_doors = [i for i in range(1,4) if((i is not winning_door) and (i is not contestant_door))]
    opened_door = remaining_doors[randint(0,len(remaining_doors))] # the door opened by the host
    return (winning_door, contestant_door, opened_door)

# simulates the experiment for num_trial trials
# returns the probability of winning by switching
def monty_hall_simulation(num_trials = 10000):
    num_wins = 0
    for i in range(num_trials):
        (winning_door, contestant_door, opened_door) = monty_hall_single_trial()
        if winning_door != contestant_door:
            num_wins += 1 # contestant wins by switching
    return (num_wins / num_trials)
```

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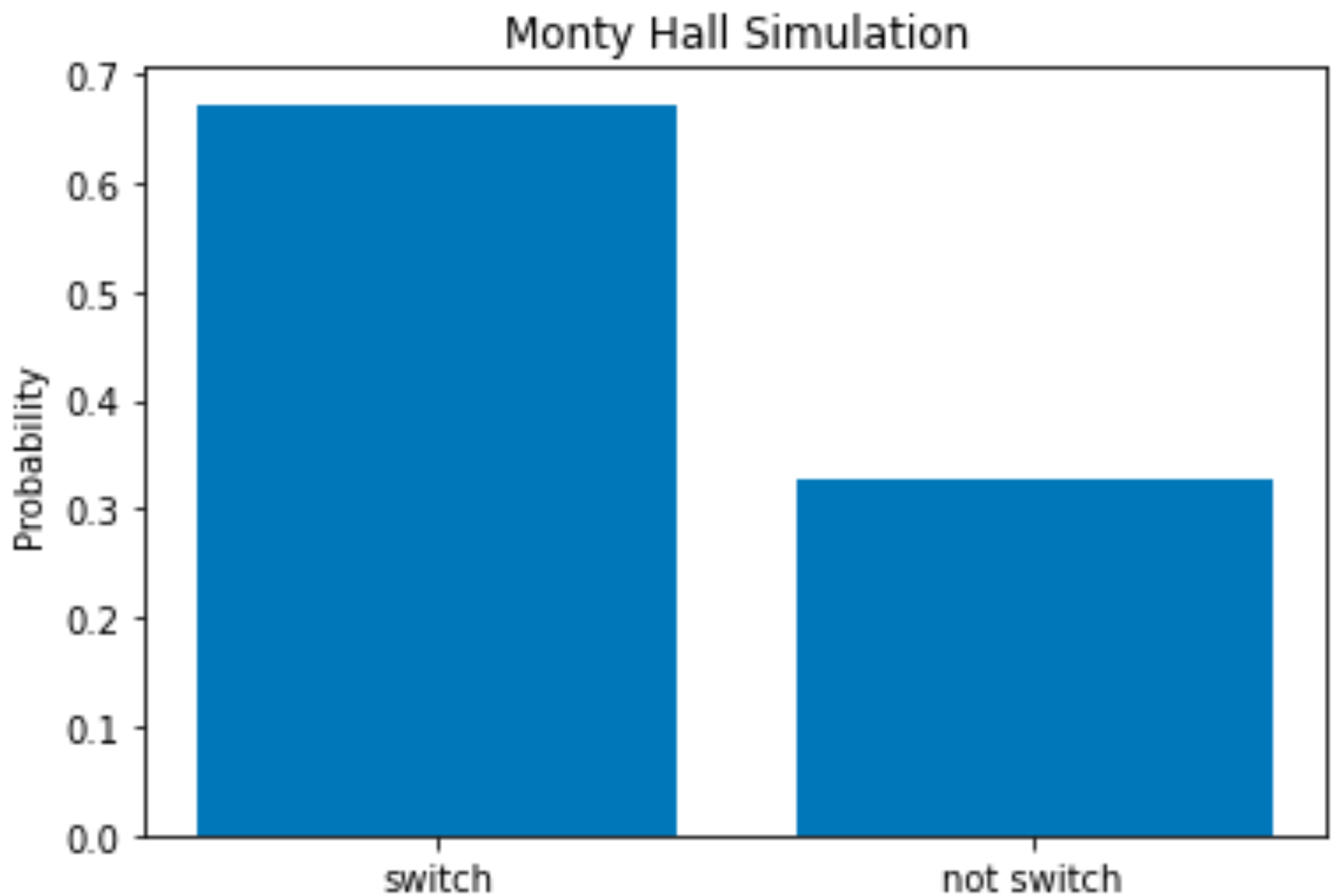
PUTTING IT TO THE TEST



10000 trials

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SUPPORT & VALIDATION

Our class, with unbridled enthusiasm, is proud to announce that our data support your position. Thank you so much for your faith in Americas educators to solve this.

–Jackie Charles, Henry Grady Elementary
Tampa, Florida

My class had a great time watching your theory come to life. I wish you could have been here to witness it. Their joy is what makes teaching worthwhile.

–Pat Pascoli, Park View School
Wheeling, West Virginia

Seven groups worked on the probability problem. The numbers were impressive, and the students were astounded.

–R. Burrichter, Webster Elementary School
St. Paul, Minnesota

The best part was seeing the looks on the students' faces as their numbers were tallied. The results were thrilling!

–Patricia Robinson, Ridge High School
Basking Ridge, New Jersey

https://www.math.ucdavis.edu/~swalcott/Monty_Hall_17Cs2018.pdf

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CONDESCENDING BACKLASH FROM ACADEMIA

> *You blew it, and you blew it big! Since you seem to have difficulty grasping the basic principle at work here, I'll explain. After the host reveals a goat, you now have a one-in-two chance of being correct. Whether you change your selection or not, the odds are the same. There is enough mathematical illiteracy in this country, and we don't need the world's highest IQ propagating more. Shame!*
> ***Scott Smith, Ph.D.***
> ***University of Florida***

> *May I suggest that you obtain and refer to a standard textbook on probability before you try to answer a question of this type again?*
> ***Charles Reid, Ph.D.***
> ***University of Florida***

> *You are utterly incorrect about the game show question, and I hope this controversy will call some public attention to the serious national crisis in mathematical education. If you can admit your error, you will have contributed constructively towards the solution of a deplorable situation. How many irate mathematicians are needed to get you to change your mind?*
> ***E. Ray Bobo, Ph.D.***
> ***Georgetown University***

> *You made a mistake, but look at the positive side. If all those Ph.D.'s were wrong, the country would be in some very serious trouble.*
> ***Everett Harman, Ph.D.***
> ***U.S. Army Research Institute***

CONDESCENDING BACKLASH FROM ACADEMIA

You blew it, and you blew it big! Since you seem to have difficulty grasping the basic principle at work here, I'll explain. After the host reveals a goat, you now have a one-in-

Takeaways: probability is hard, even for experts.

- We need to take a rigorous, proof-based approach to obtain the right answers.
- Tree diagrams can help us make sense of tricky situations (we will use one to understand Monty Hall).
- Doing examples by hand or with code can help a lot.

E. Ray Bobo, Ph.D.
Georgetown University

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Everett Harman, Ph.D.
U.S. Army Research Institute

ASIDE: LARGE LANGUAGE MODELS AND YOU

- You may be tempted to use ChatGPT or another LLM to help you with your homework.
- This is against the code of conduct. But even if it was not, this is not in your interest.
  - Homework is only 25% of the grade in this course.
  - Data suggests that using LLMs to solve the homework will result in a lower overall course grade, even if it improves your homework score.
  - *You need to show your brain that you can solve these problems without help so that it can do so during an exam.*

GEN

Generative AI Can Harm Learning

Hamsa Bastani,[1*] Osbert Bastani,[2*] Alp Sungu,[1*†]
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[1]Operations, Information and Decisions, University of Pennsylvania
[2]Computer and Information Science, University of Pennsylvania
[3]Wharton AI & Analytics, University of Pennsylvania
[4]Budapest British International School

- University of Pennsylvania researchers compared the progress of nearly 1,000 students by evaluating how they performed on homework problems, then on exams.
- Group A was allowed to use ChatGPT to solve their homework problems, and Group B was not.
- Group B, who did not use ChatGPT, <u>outscored</u> Group A on exams.
- In fact, students using ChatGPT scored 17% worse on exams.

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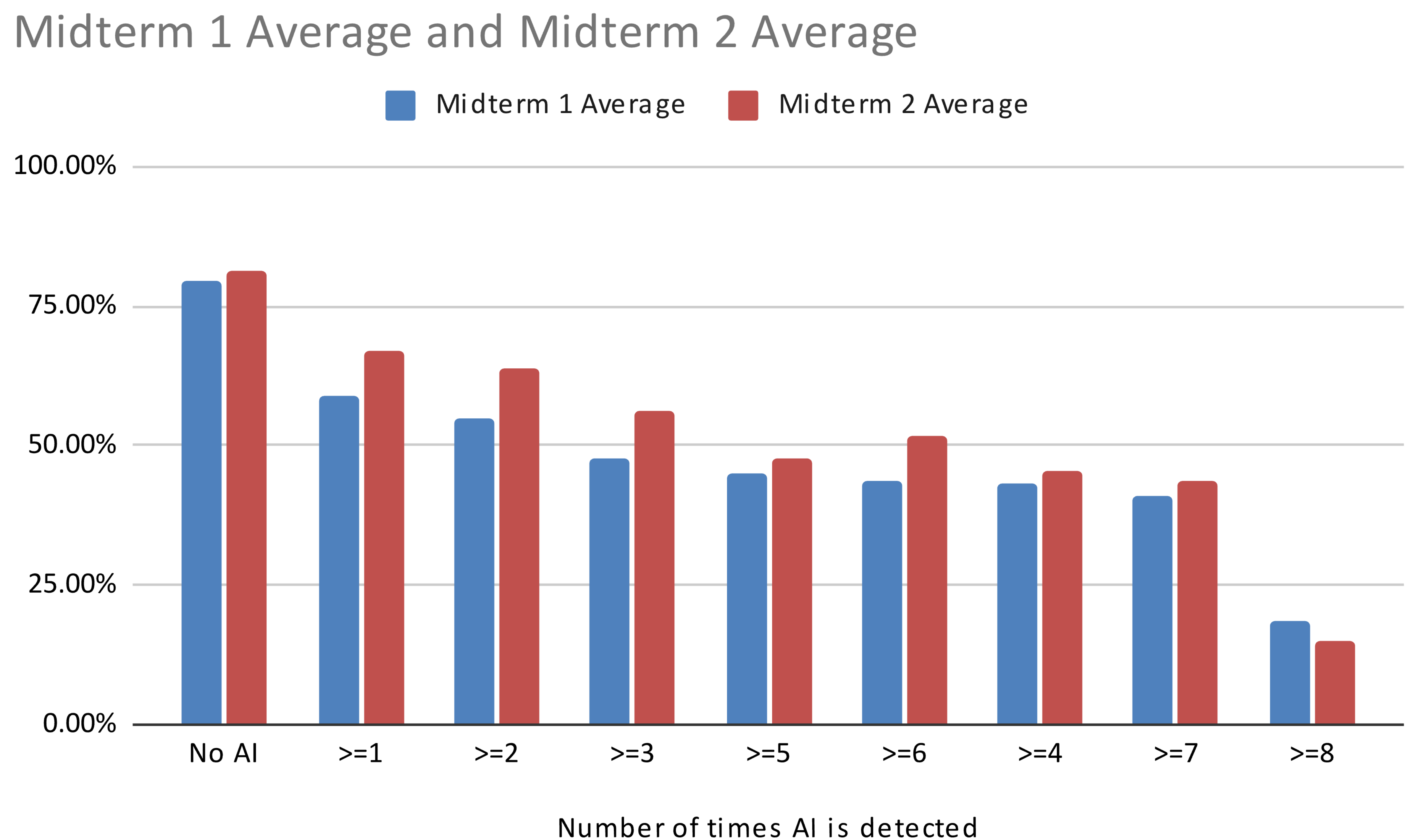
AI DETECTION VS MIDTERM AVERAGE



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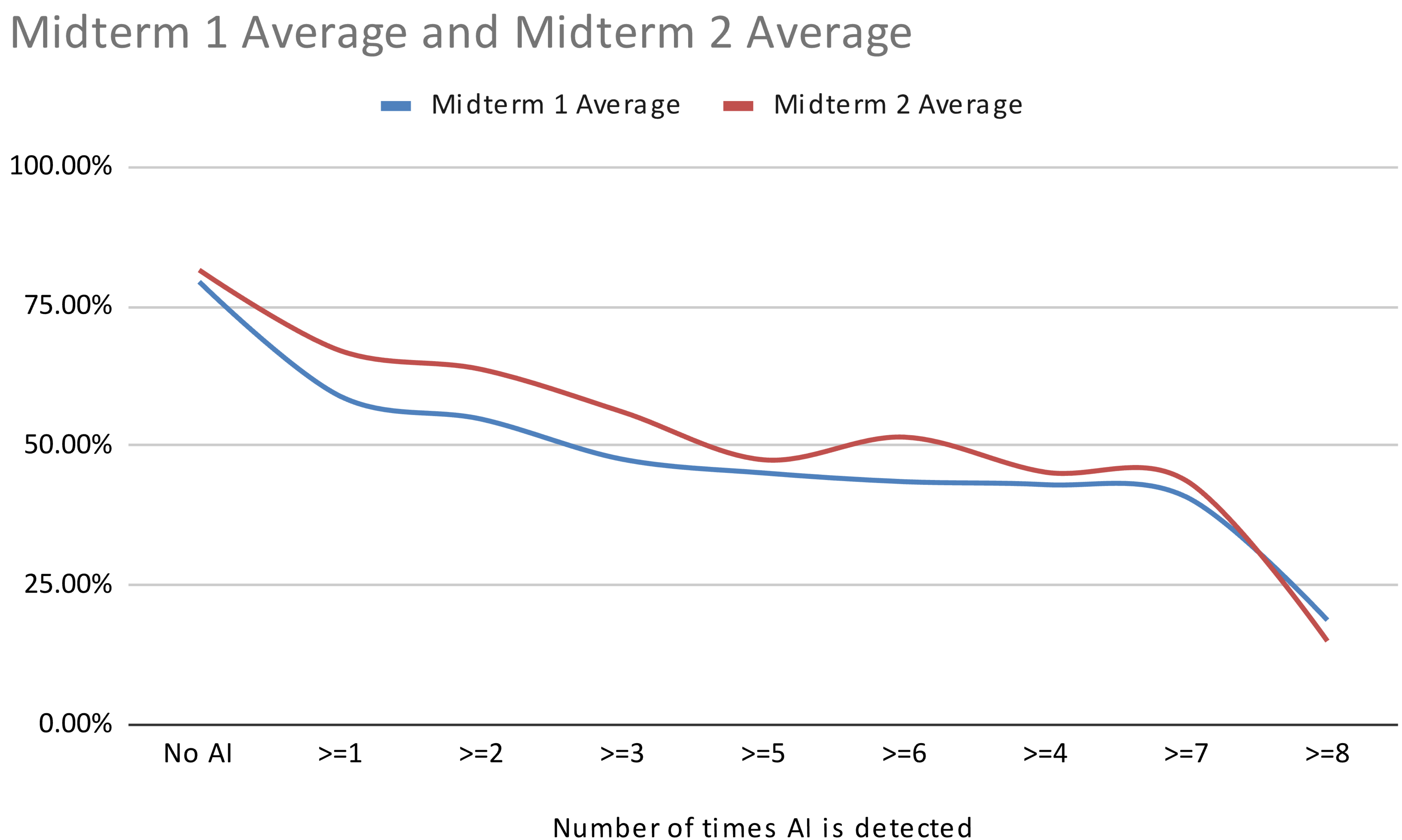
AI DETECTION VS MIDTERM AVERAGE



AI D

You need to show your brain that you can solve these problems without help so that it can also do so during an exam.